

BEGINNING OF TRANSMISSION

State: unpublished / pending pairing
Pairing: pending — unresolved: - GitHub release URL: pending - PDF SHA-256: pending

Release metadata

Field	Value
Title	Active Inference Multi-Track Exemplar
Version	0.3.0
Concept DOI	10.5281/zenodo.20417021
Version DOI	10.5281/zenodo.20420352
GitHub	docxology/template_active_inference
Zenodo	https://zenodo.org/records/20417021
SHA-256	pending
SHA-512	pending

How to verify

- Scan **Integrity** QR and compare the embedded SHA-256 prefix to the table above.
- Scan **Zenodo** / **GitHub** QR codes and confirm they resolve to this release pairing.
- Full hashes and structured fields: `../data/transmission_manifest.json`.



Figure 1: Integrity QR strip

Structured manifest: `../data/transmission_manifest.json`

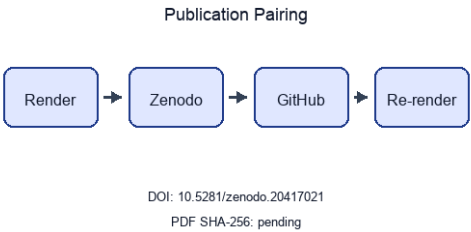


Figure 2: Publication pairing flow

Stego: off | overlays text | barcodes on | XMP on | manifest on → `./secure_run.sh`

Active Inference Multi-Track Exemplar

Sheaf-Composed Manuscript with pymdp Sophisticated Inference

Daniel Ari Friedman
Active Inference Institute
`daniel@activeinference.institute`
ORCID: [0000-0001-6232-9096](https://orcid.org/0000-0001-6232-9096)
DOI: [10.5281/zenodo.20417021](https://doi.org/10.5281/zenodo.20417021)

June 2, 2026

Contents

1 Sheaf Track Coverage	3
1.1 Introduction	3
2 Abstract	4
Introduction	5
3 Motivation and scope	5
3.1 Scientific scope	5
3.2 Manuscript structure	5
4 Contributions	6
4.1 Scientific contributions	6
4.1.1 Ontology bindings	6
Methods	7
5 Bernoulli–Ising analytical model	7
5.0.1 Ontology bindings	7
6 pymdp simulation harness	8
6.0.1 Ontology bindings	8
7 Lean formalization boundary	10
7.0.1 Proof extraction track	10
8 Sheaf composition	11
8.1 Compose contract	11
8.2 Coverage and figures	11
8.3 Compose commands	11
8.4 Law verification	11
8.4.1 Base poset and presheaf	11
8.4.2 Verified sheaf laws	11
8.4.3 Scope (what is and is not claimed)	12
8.4.4 Artifact diffoscope track	13
8.4.5 Artifact license track	13
8.5 Sheaf fragment track registry	13
8.6 IMRAD binding matrix	14
8.7 Section-track status	16
8.8 Track status	16
8.9 Render and logging summary	17
8.10 Evidence crosswalk	17
8.11 Artifact producer graph	18
8.12 Semantic gluing restrictions	20
8.13 Track improvement scope	21
Results	22
9 Mutual-information parameter sweep	22
10 Free-energy decomposition	23
11 T-maze active-inference rollout	24
12 Validation invariants	26
12.0.1 State-space catalog track	26
12.0.2 Causal ablation track	26
Discussion	27
13 Limitations and outlook	27
13.1 What this demonstrates	27
13.2 Limitations	27
13.3 Sheaf audit and outlook	27
13.3.1 Ontology bindings	27
13.3.2 Release notes evidence track	27

Appendix	28
14 Appendix: full track coverage	28
14.0.1 Appendix track: artifact diffoscope	28
14.0.2 Appendix track: artifact license	28
14.0.3 Appendix track: proof extraction	30
14.0.4 Appendix track: state-space catalog	30
14.0.5 Appendix track: causal ablation	30
14.0.6 Ontology bindings	30
14.0.7 Appendix track: release notes evidence	30
15 Conclusion	31
16 References	32

1 Sheaf Track Coverage

This page summarizes which **sheaf fragment tracks** are bound for each IMRAD row in `manuscript/sheaf/manifest.yaml1`. The matrix is regenerated at compose time.

Totals: 90 present / 90 bound / 0 missing (gray).

Color	Meaning
Black	Track present (bound and fragment exists)
White	Absent (not bound for this row)
Gray	Missing (bound but fragment file absent)

1.1 Introduction

- **Introduction** (*group*)
- **Motivation and scope**
- **Contributions** ## Methods
- **Methods** (*group*)
- **Bernoulli–Ising analytical model**
- **pymdp simulation harness**
- **Lean formalization boundary**
- **Sheaf composition** ## Results
- **Results** (*group*)
- **Mutual-information parameter sweep**
- **Free-energy decomposition**
- **T-maze active-inference rollout**
- **Validation invariants** ## Discussion
- **Discussion** (*group*)
- **Limitations and outlook** ## Appendix
- **Appendix** (*group*)
- **Appendix: full track coverage**

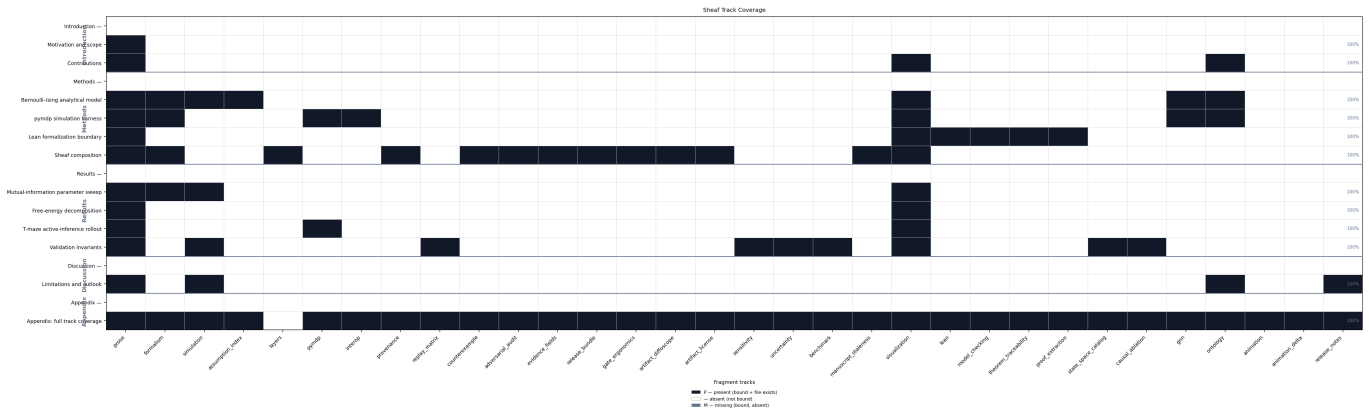


Figure 3: Sheaf track coverage matrix: 17 IMRAD rows $\times$ 32 fragment columns. Black = present (P), white = absent (—), gray = missing (M). Counts: 90 present / 90 bound / 0 missing.

Appendix row `16_appendix_full_sheaf.md` binds 31 fragment track types as a composability proof (registry defines 32 types; optional `layers` is methods-only).

2 Abstract

We study a minimal Active Inference stack on toy models: a Bernoulli–Ising analytical oracle, a pymdp T-maze rollout, and a sheaf-indexed compose contract that binds 32 fragment tracks into 12 flat IMRAD sections. The methodological contribution is a discipline rather than a domain finding: every reported number is hydrated from a generated artifact and every cross-track claim is machine-checked before rendering, so no figure or statistic can drift from the artifact that produced it — 6 sheaf axioms are verified before composition and 22 negative controls keep each failure path live. Claims are limited to those models and their generated artifacts.

sec. 1 reports a 17-row coverage matrix (5 IMRAD group headers) regenerated from the live manifest at compose time. sec. 6 documents the T-maze harness aligned with [pymdp sophisticated_inference examples](#).

sec. 12 records 12 / 12 invariant checks passed. SI planning horizon: 2 steps. Sweep RMSE 0 nats bounds analytical–empirical agreement on the coupling grid.

Introduction

3 Motivation and scope

3.1 Scientific scope

This manuscript couples three tracks on toy Active Inference models: a Bernoulli–Ising analytical oracle, a pymdp T-maze rollout, and a sheaf-indexed assembly contract that binds 32 optional fragment tracks under an IMRAD outline. The scientific claims stay within those models and their generated artifacts; they are not empirical statements about biological agents.

3.2 Manuscript structure

Three **scientific tracks** (analytical, pymdp, sheaf composition) map onto 32 **composable fragment types** and 29 pipeline gates (fig. 4). sec. 1 summarizes which fragment tracks bind to each manifest row. sec. 8 documents the compose pipeline, coverage semantics (eq. 3), and strict validation gates.

The pymdp track follows the **pymdp sophisticated_inference examples** with a minimal T-maze and planning horizon **policy_len** = 2. Other sections cite sec. 6 instead of repeating that reference.

4 Contributions

4.1 Scientific contributions

1. **Analytical oracle** (sec. 5): closed-form mutual information and free-energy decomposition on a symmetric Bernoulli–Ising toy with an independent exact-recomputation cross-check (sec. 9, sec. 10).
2. **Active-inference harness** (sec. 6): deterministic pymdp T-maze rollout — default `state_inference` belief filtering, with sophisticated expected-free-energy policy inference selectable via `mode: policy_inference` — with logged beliefs, actions, and merged invariant gates (sec. 11, sec. 12).
3. **Sheaf-indexed composition** (sec. 8): 32 optional fragment types bind to 17 manifest rows under eq. 3, with a 31-track appendix composability proof (sec. 14).

fig. 4 maps the three scientific tracks to 29 pipeline gates and 32 composable fragment renderers. Measured invariant checks: 12 / 12 passed.

Ontology-facing symbols are checked per model: the Bernoulli toy binds `pi1`, `pi2`, `J`, `gamma`, and `q_joint`, while the SI T-maze binds `location`, `observation`, `policy`, and `belief_entropy` to `HiddenState`, `ObservationLikelihood`, `PolicyPosterior`, and `BeliefEntropy` (fig. 6, sec. 6).

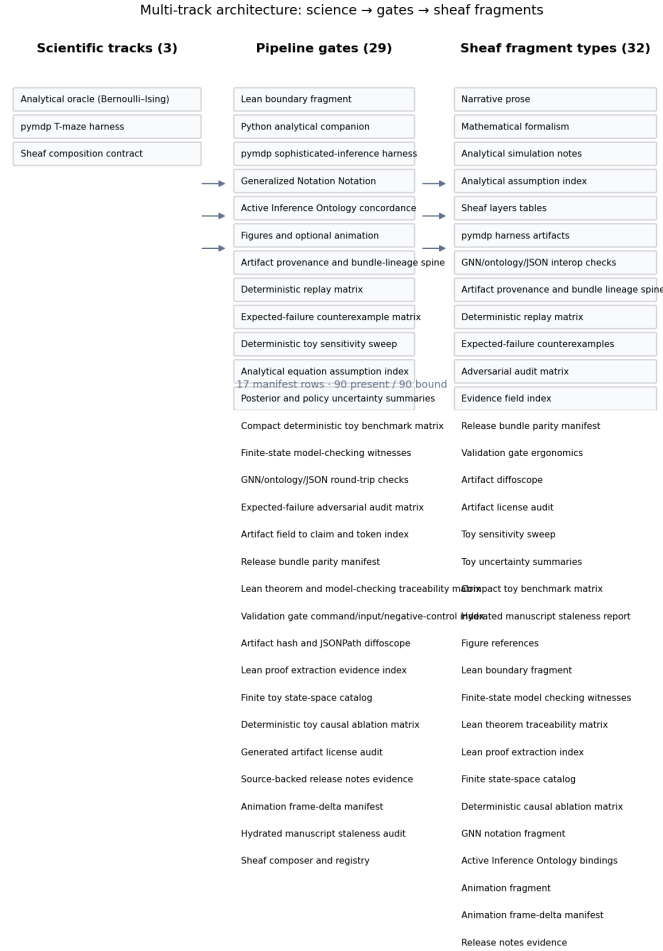


Figure 4: Multi-track architecture: analytical, pymdp, and sheaf composition lanes mapped to 29 pipeline gates and 32 composable fragment types.

4.1.1 Ontology bindings

- `expected_free_energy` → `ExpectedFreeEnergy`
- `location` → `HiddenState`
- `observation` → `ObservationLikelihood`
- `policy` → `PolicyPosterior`

Methods

5 Bernoulli–Ising analytical model

We study a minimal **K=2 Bernoulli / Ising** coupling as the analytical companion to multi-track verification. The entangled joint eq. 1 yields closed-form mutual information $I(\lambda)$, which serves as an oracle for an independent exact recomputation (via total correlation) and GNN round-trips in sec. 9 (fig. 6).

Measured sweep grid points: 21. Invariants passed: 12 / 12.

The entangled joint over binary policies satisfies

$$q_\lambda(\pi) \propto E(\pi) \exp(\lambda J(\pi)), \quad (1)$$

with symmetric Ising coupling J and deformation parameter λ . Let $\sigma(\lambda) = q_\lambda(\pi_1 = \pi_2)$ be the probability that the two streams agree (the diagonal mass of the 2×2 joint); by symmetry both marginals are uniform. With binary entropy $H_b(p) = -p \log p - (1-p) \log(1-p)$ in nats, the joint entropy is $H(q_\lambda) = \log 2 + H_b(\sigma(\lambda))$ while each marginal contributes $\log 2$, so the mutual information is

$$I(\lambda) = \sum_k H(q_k) - H(q_\lambda) = \log 2 - H_b(\sigma(\lambda)),$$

vanishing at $\lambda = 0$ ($\sigma = \frac{1}{2}$, independent streams) and saturating at $\log 2$ as $\lambda \rightarrow \infty$ ($\sigma \rightarrow 1$, perfectly entangled). These symbols are the rows of `analytical_assumption_index.json`, so the derivation is auditable rather than asserted.

The analytical track writes a parameter sweep comparing closed-form mutual information with an independent exact recomputation of it (via total correlation) across $\lambda \in [0, 4]$ on 21 grid points (sec. 9, fig. 5).

The `assumption_index` fragment makes the analytical equations inspectable as a generated artifact instead of relying on prose labels. `output/data/analytical_assumption_index.json` indexes 6 finite-model equation identifiers and 6 rows; the hydrated pass flag is `true`.

The index is deliberately narrow. It covers the Bernoulli-Ising toy equations, their finite binary state assumptions, and the generated artifacts that test the same symbols. Any missing equation identifier or empty assumption list fails the toy-sweep validation gate.

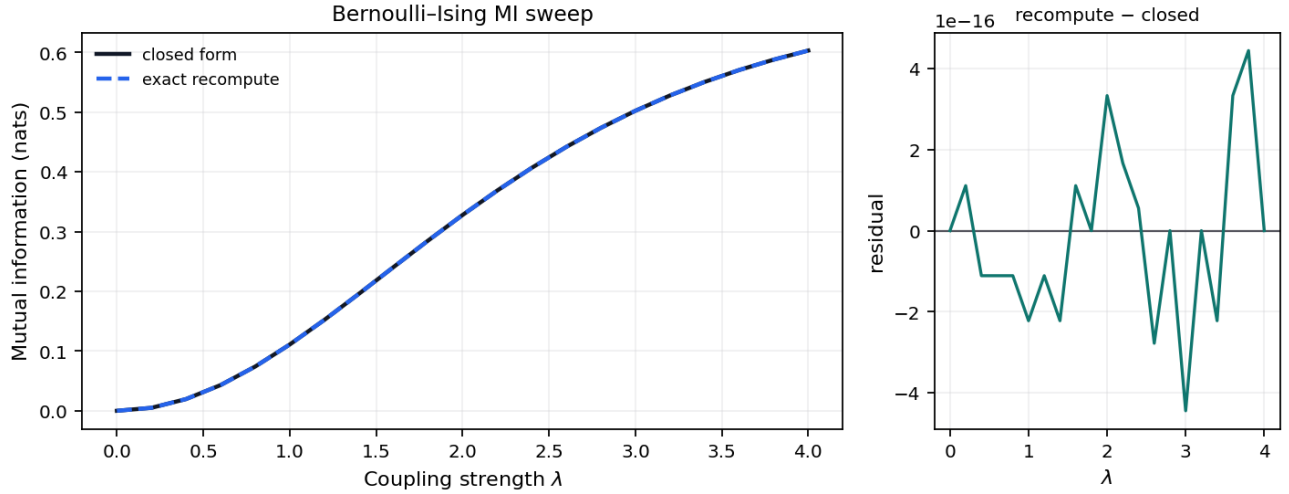


Figure 5: Closed-form $I(\lambda)$ and an independent exact recomputation via total correlation for the symmetric Bernoulli-Ising toy across 21 grid points up to $\lambda_{\max} = 4$; grid maximum 0.6031 nats. Both estimators are deterministic (no sampling), so the right panel is a cross-implementation agreement check (max residual 0 nats), not a sampling residual.

The Bernoulli toy is declared in `gnn/bernoulli_toy.gnn.md` (GNN v1.1). fig. 6 links GNN variables to Active Inference Ontology terms bound in the analytical ontology fragment; round-trip parity is checked before render.

Measured MI and sweep artifacts in sec. 9 ground the same symbol map used in the concordance diagram.

5.0.1 Ontology bindings

- `E1` → `Stream1HabitPrior`
- `E2` → `Stream2HabitPrior`
- `J` → `CrossStreamCouplingPotential`
- `gamma` → `SophisticationWeight`
- `lam` → `EntanglementDeformationParameter`
- `pi1` → `Stream1PolicyVector`
- `pi2` → `Stream2PolicyVector`
- `q_joint` → `EntangledJointPosterior`

Analytical symbol		GNN $\leftrightarrow$ ontology concordance (GNN v1.1)		Ontology term
		GNN variable		
π^1	_____	π_1	_____	Stream1PolicyVector
π^2	_____	π_2	_____	Stream2PolicyVector
E1	_____	E1	_____	Stream1HabitPrior
E2	_____	E2	_____	Stream2HabitPrior
J	_____	J	_____	CrossStreamCouplingPotential
λ	_____	λ	_____	EntanglementDeformationParameter
γ	_____	γ	_____	SophisticationWeight
q	_____	q_{joint}	_____	EntangledJointPosterior

Figure 6: GNN $\leftrightarrow$ ontology concordance for the Bernoulli–Ising toy (GNN v1.1).

6 pymdp simulation harness

Sophisticated inference (planning horizon). This section documents a **pymdp state-inference harness** on a minimal T-maze (fig. 7) with planning horizon `policy_len = 2`. Default mode is `state_inference` (T-maze rollout via `simulate_si_tmaze.py`). The Agent constructs a multi-step policy set (`num_policies` logged in SI artifacts); per-step **belief entropy** is aggregated as `mean_belief_entropy`. Selecting mode: `policy_inference` enables expected-free-energy policy selection; on the default minimal T-maze (2 states / 2 observations / 2 actions, horizon 2) this yields a constant-action policy that does not reach the goal — the toy is deliberately too small to exercise sophisticated inference’s advantage, so richer state spaces and longer horizons are needed for it to differ from belief filtering.

Graph-world artifacts are deterministic extension outputs declared in `tracks.yaml` `extension_tracks.graph_world`. For the reference workflow, see sec. 3; measured rollouts appear in sec. 11.

Mean belief entropy across steps: 0.3251.

The comparison artifact `output/data/si_policy_comparison.json` runs both `state_inference` and `policy_inference` over the declared toy horizons and seeds without replacing the default rollout. It records 4 deterministic comparison rows, complete-grid flag 1, and 3 goal-reaching rows under the toy transition model.

Agent construction is audited in `output/reports/pymdp_runtime_diagnostics.json`: 4 constructions, 4 known third-party JAX static-array warnings, and 0 unexpected warnings. Policy posterior evidence is written separately to `output/data/pymdp_policy_posterior_grid.json` with 10 rows and normalized-posterior flag 1.

The graph-world extension is deterministic: `simulate_si_graph_world.py` writes `si_graph_world_summary.json` and `si_graph_world_trace.json` for a four-node graph-world path. The regenerated summary reports 4 nodes, 4 steps, and goal-reached flag 1. The topology-trace extension records 4 topology traces with agreement flag 1.

Given generative matrices A, B, C, D , pymdp computes state beliefs $q(s)$ via variational inference (`infer_states`). The Agent is configured with planning horizon $H = 2$, which defines the **policy depth** used when constructing candidate policies (logged as `num_policies` in the SI summary artifact; see sec. 11).

The default harness records belief entropy per step; extending to full expected-free-energy policy selection (`infer_policies`) is documented as a follow-on track in sec. 13.

SI artifacts (summary, trace, optional JSONL log) record step count, actions, observations, and belief entropy for sec. 11. Steps recorded: 2.

The **interop** fragment treats the GNN files, JSON views, and ontology bindings as a round-trip contract rather than parallel documentation. `output/data/interop_roundtrip_report.json` records 5 deterministic checks; the manuscript only claims losslessness when `true` is true.

The stricter lint artifacts are adjacent evidence, not new model claims: `output/data/gnn_roundtrip_report.json`, `output/reports/gnn_lint_report.json`, `output/data/ontology_alias_index.json`, and `output/data/ontology_profile_matrix.json` must agree before the interop row passes. A missing GNN variable, duplicate ontology alias, dropped JSON field, shape diff, or dtype diff is therefore a validation failure before rendering.

See `gnn/si_tmaze.gnn.md` for a GNN view of the T-maze hidden state, observation, and policy variables with ontology bindings.

6.0.1 Ontology bindings

- `belief_entropy` $\rightarrow$ `BeliefEntropy`

Minimal T-maze POMDP schematic

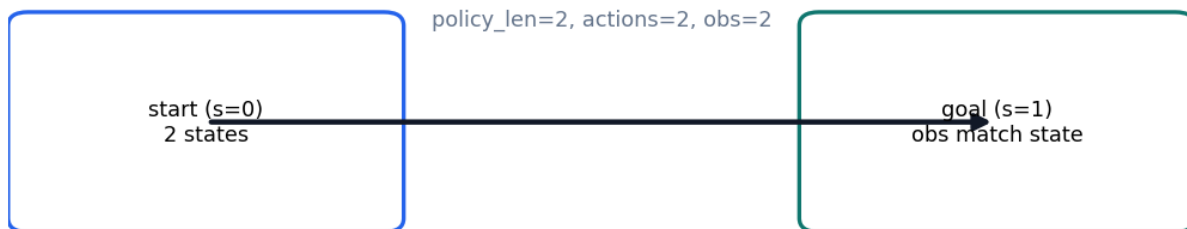


Figure 7: T-maze generative model schematic (2-step policy horizon, state_inference mode).

- loc $\rightarrow$ **HiddenState**
- obs $\rightarrow$ **ObservationLikelihood**
- pi $\rightarrow$ **PolicyPosterior**

7 Lean formalization boundary

The Lean track provides minimal boundary witnesses checked by `lake build` under `lean/TemplateActiveInference/`. fig. 8 summarizes proved versus deferred statements; fragments cite theorem names without duplicating proof scripts in prose.

Horizon witnesses link back to the analytical toy (sec. 5) and the pymdp planning depth (sec. 6).

Lean formalization boundary status

Module	Kind	Name	Status
TemplateActiveInference.Bernoul	def	isingCouplingEntries	proved
TemplateActiveInference.Bernoul	def	isingCouplingSum	proved
TemplateActiveInference.Bernoul	theorem	ising_coupling_sum_zero	proved
TemplateActiveInference.Sophist	def	defaultPolicyLen	proved
TemplateActiveInference.Sophist	theorem	sophisticated_requires_horizon	proved
TemplateActiveInference.Sophist	def	tmazeStep	proved
TemplateActiveInference.Sophist	theorem	tmaze_two_forward_steps_reach	proved
TemplateActiveInference.Sophist	theorem	tmaze_goal_absorbing	proved
TemplateActiveInference.Sophist	def	graphWorldStep	proved
TemplateActiveInference.Sophist	theorem	graph_world_three_steps_reach	proved
TemplateActiveInference.Sophist	def	branchGraphWorldStep	proved
TemplateActiveInference.Sophist	theorem	branch_graph_world_three_steps	proved
TemplateActiveInference.Sophist	def	loopGraphWorldStep	proved
TemplateActiveInference.Sophist	theorem	loop_graph_world_four_steps_rea	proved
TemplateActiveInference.Sophist	def	diamondGraphWorldStep	proved
TemplateActiveInference.Sophist	theorem	diamond_graph_world_four_steps	proved
TemplateActiveInference.Sophist	def	finitePolicies	proved
TemplateActiveInference.Sophist	theorem	policy_enumeration_contains_for	proved
TemplateActiveInference.Sophist	def	twoStateBeliefWeights	proved
TemplateActiveInference.Sophist	theorem	two_state_belief_weights_sum_to	proved
TemplateActiveInference.Sophist	def	twoPolicyPosteriorWeights	proved
TemplateActiveInference.Sophist	theorem	two_policy_posterior_weights_sur	proved

Figure 8: Lean formalization boundary: module witnesses checked by lake build.

Lean module `TemplateActiveInference.SophisticatedInference` declares the planning-horizon parameter `defaultPolicyLen` and finite T-maze boundary witnesses: `sophisticated_requires_horizon : defaultPolicyLen > 1`, `tmaze_two_forward_steps_reach_goal`, and `tmaze_goal_absorbing`. It also contains constructive finite witnesses for graph-world reachability, finite policy enumeration, two-state belief weights, and two-policy posterior weights. These theorems formalize small finite boundaries shared with generated artifacts; they do *not* prove that the toy policy posterior is a general model of sophisticated inference. Axioms are audited with `#print axioms` (the gate whitelists only `propext`, `Classical.choice`, `Quot.sound`); see the Lean track gate.

Build via `lake build` under `lean/`.

The `model_checking` fragment complements Lean with finite exhaustive witnesses. `output/reports/model_checking_witnesses.json` records 12 toy-state witnesses and reports `true` only when no counterexample is found in the enumerated state/action space.

This is deliberately narrower than a semantic proof of all Active Inference programs. It checks the finite T-maze and graph-world boundary objects used by this manuscript and exposes the witness inventory to the same artifact and claim gates as the Lean theorem inventory. The Lean graph-world inventory witnesses 4 generated toy topology ids, with all-topologies-witnessed flag `true`; theorem traceability contributes 11 linked rows.

The `theorem_traceability` fragment binds Lean theorem inventory rows to finite model-checking witnesses, manuscript claims, and evidence fields. `output/data/theorem_traceability_matrix.json` records 11 traceability rows and passes only when every theorem row is linked (`true`).

7.0.1 Proof extraction track

The `proof_extraction` track extracts Lean theorem statements and proof-source metadata into `output/data/proof_extraction_index.json`. The index currently contains 10 extracted theorem rows, with constructive-token status `true`.

8 Sheaf composition

8.1 Compose contract

Each manifest row in `manuscript/sheaf/manifest.yaml` binds fragment tracks from `manuscript/sheaf/tracks.yaml`. A track supplies a renderer, compose order, label, and optional flag; the composer flattens the binding set into one Markdown section for PDF and web output.

The operational claim is auditable binding: analytical, simulation, pymdp, visualization, Lean, GNN, ontology, and optional media fragments attach to each IMRAD row under eq. 3 (**P** present, — unbound, **M** missing).

8.2 Coverage and figures

fig. 9 summarizes 32 fragment types and their IMRAD bindings. Generated tables below list every track definition and section×track binding at compose time.

8.3 Compose commands

```
uv run python scripts/compose_manuscript.py
uv run python scripts/compose_manuscript.py --validate-only --strict
```

Each run emits `output/data/sheaf_coverage_matrix.json` and regenerates coverage artifacts. Partial compose (`--section`) is draft-only; the matrix always reflects the full manifest. Coverage totals appear on sec. 1; discussion scope is in sec. 13.

8.4 Law verification

`--validate-only --strict` runs the structural gate before any fragment is glued. Beyond per-cell coverage, it invokes the sheaf-law oracle (`verify_sheaf_laws`, `src/manuscript/sheaf/laws.py`), which checks 6 axioms — poset, presheaf functoriality, separation, gluing, typing, and compositionality — and reports 6/6 satisfied for the current manifest. A violation is raised as an error-level issue and aborts the build, so a malformed manifest (a section colliding on an output file, an off-chain block, a mistyped fragment, a fragment shared between sections) can never compose. The formal statements are in the formalism block below; the negative-control suite (`tests/test_sheaf_laws.py`) proves each check is falsifiable.

The semantic layer is separate from those structural laws. `output/data/sheaf_gluing_certificate.json` records cross-track symbols, typed claim evidence, artifact sources, and manuscript-variable restrictions; validation fails when the analytical, pymdp, GNN, ontology, Lean, visualization, or manuscript tracks disagree about a shared symbol or measured claim. fig. 10 renders this gluing graph: the configured producers, the generated evidence artifacts, and the validation consumers that read each shared symbol.

8.4.1 Base poset and presheaf

The manuscript is modelled as a coverage sheaf over a finite base poset. Let the **base** P be the IMRAD blocks ordered as a chain,

$$\text{Introduction} \prec \text{Methods} \prec \text{Results} \prec \text{Discussion} \prec \text{Appendix}, \quad (2)$$

with, in each block, a *group* node above its *section* nodes (written $G \sqsupseteq s$). P is therefore a finite poset (equivalently a finite Alexandrov space). Let $\mathcal{T}$ be the registered fragment-track set from `manuscript/sheaf/tracks.yaml`; each track $t \in \mathcal{T}$ carries a renderer $R(t)$, label $L(t)$, optional flag $O(t)$, and a strict compose-order index $\pi(t)$.

The **presheaf** $\mathcal{F}$ is a contravariant functor on $P \leftarrow \mathcal{F} \rightarrow \mathbf{Set}$ with restriction maps along $\sqsupseteq$ — assigning to each composing section s its bound fragment set $\mathcal{F}(s) = \{(t, F_s(t)) : t \text{ bound in } s\}$, where $F_s : \mathcal{T} \rightarrow \mathbf{Path}$ is the section’s partial binding map. Restriction along $G \sqsupseteq s$ is projection onto a section’s own bindings; group nodes carry the empty assignment and do not compose.

The coverage cell is

$$B(s, t) \in \{\mathbf{P}, \text{—}, \mathbf{M}\} \quad (3)$$

derived from $F_s(t)$ and filesystem existence at compose time: **P** when a bound fragment exists, — when the track is unbound for that row, and **M** when a bound path is missing. The current regenerated matrix reports 90 present / 90 bound / 0 missing cells. Registry size: $|\mathcal{T}| = 32$ types across 17 IMRAD manifest rows (5 group rows, 12 composing sections).

8.4.2 Verified sheaf laws

What makes this presheaf a *sheaf* — rather than a bare incidence table — is that the composer’s structural axioms are machine-checked. The oracle `verify_sheaf_laws` (`src/manuscript/sheaf/laws.py`) verifies 6 laws, and the regenerated build reports 6/6 satisfied:

1. **Poset**. The IMRAD blocks form the chain of eq. 2; compose order is monotone in block rank and every composing section’s block carries a group row.
2. **Presheaf (functoriality)**. Every bound track lies in $\mathcal{T}$; π is a strict total order; and each section’s resolved track order is the monotone restriction of π (an explicit `track_order` override must be a permutation of the section’s bound tracks).
3. **Separation (locality)**. The map $s \mapsto \text{output_name}(s)$ is injective over composing sections: distinct locals glue to distinct global positions, so the global section is unique.
4. **Gluing**. Compose order is a linear extension of P — each block’s rows are contiguous and strictly increasing in order — so the local fragments glue to a unique global manuscript in which every composing section appears exactly once.

5. **Typing.** Each binding $(t, F_s(t))$ is well-typed: $R(t)$ is a registered renderer and the fragment suffix lies in $R(t)$'s accepted suffix set. Generated renderers (`section_figures`, `layers_report`) synthesize their body and are explicitly type-exempt.
6. **Compositionality.** Every fragment file is private to one section (no path is bound twice), so global composition is the coproduct of the per-section bodies and is independent of inclusion order.

Each law is paired with a negative control in `tests/test_sheaf_laws.py` — a single mutation that breaks the law and is proven to be caught — so the gate binds the laws' *content*, not merely their shape. Under `--strict`, any violation is surfaced as an error-level manifest issue and aborts composition.

8.4.3 Scope (what is and is not claimed)

These laws verify the sheaf *axioms* on a finite base poset. They do **not** compute sheaf *cohomology* (H^0/H^1 , Čech complexes, derived functors); “sheaf” here names the verified separation-and-gluing structure of a multi-track coverage assignment, not a cohomological invariant. Formal track definitions and section×track bindings appear in the generated tables below.

Semantic gluing then checks agreement of the glued content: coverage counts, manuscript variables, typed claim predicates, pymdp mode/hash, Bernoulli GNN ontology, and SI T-maze GNN ontology. This certificate is a content-level audit over the same base, not an additional topological law.

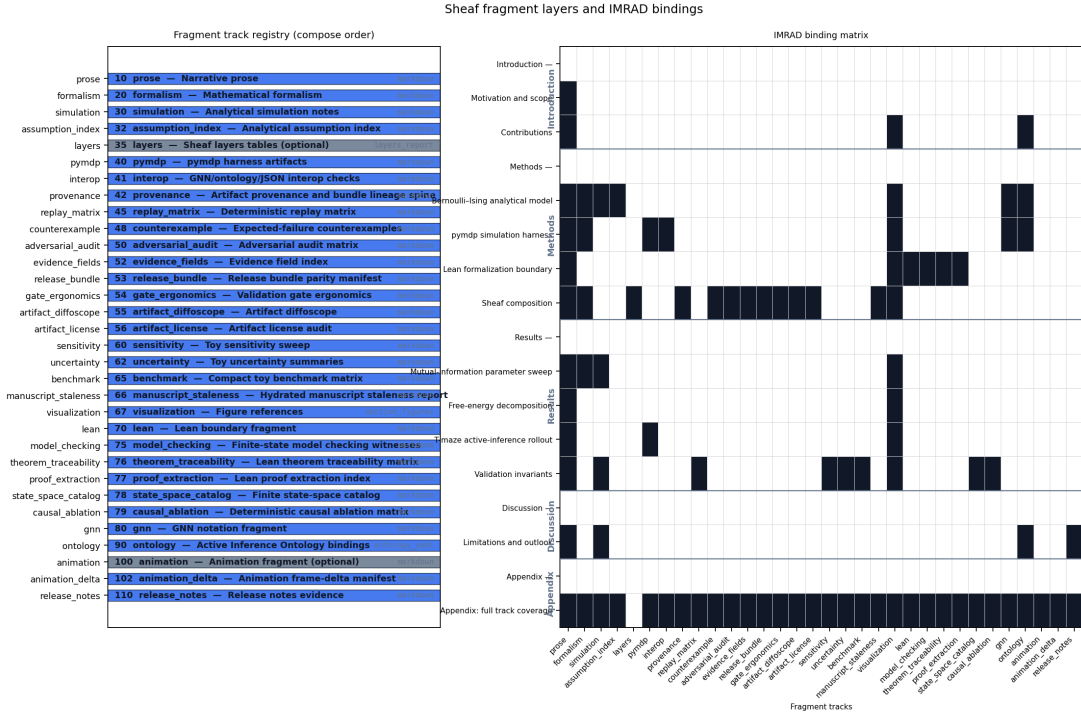


Figure 9: Sheaf layers overview: registry stack (compose order, renderer ids) and IMRAD binding heatmap for 32 fragment types across 17 manifest rows (90 present / 90 bound / 0 missing).

The `provenance` fragment makes artifact lineage a live canonical sheaf track. The configured producer `generate_sheaf_tracks.py` writes `output/data/artifact_provenance.json`, which hashes 70 required toy artifacts and records producer scripts, source commit, deterministic seed fields, config digests, and 5 artifact bundles. Publication claims that depend on generated files must be traceable to this lineage table or to a narrower artifact-specific certificate.

The `provenance` claim is intentionally limited: every listed artifact exists, has a SHA-256 digest or an explicit cycle exclusion, is produced by a configured analysis script, and carries seed/config provenance (70 seeded rows; all seeded flag `true`; bundle-complete flag `true`). A changed file, missing producer, or stale saved digest is a validation failure, not a prose warning.

The `counterexample` fragment records expected-failure fixtures as first-class evidence. `output/reports/counterexample_matrix.json` lists 22 negative controls that intentionally mutate ontology mappings, semantic certificates, graph-world trace agreement, typed claim evidence, replay rows, release parity, and provenance hashes.

The matrix is not an empirical result. It is a falsifiability ledger: each row names the gate that must fail and the test that proves the failure path remains live.

Semantic sheaf gluing dependency graph

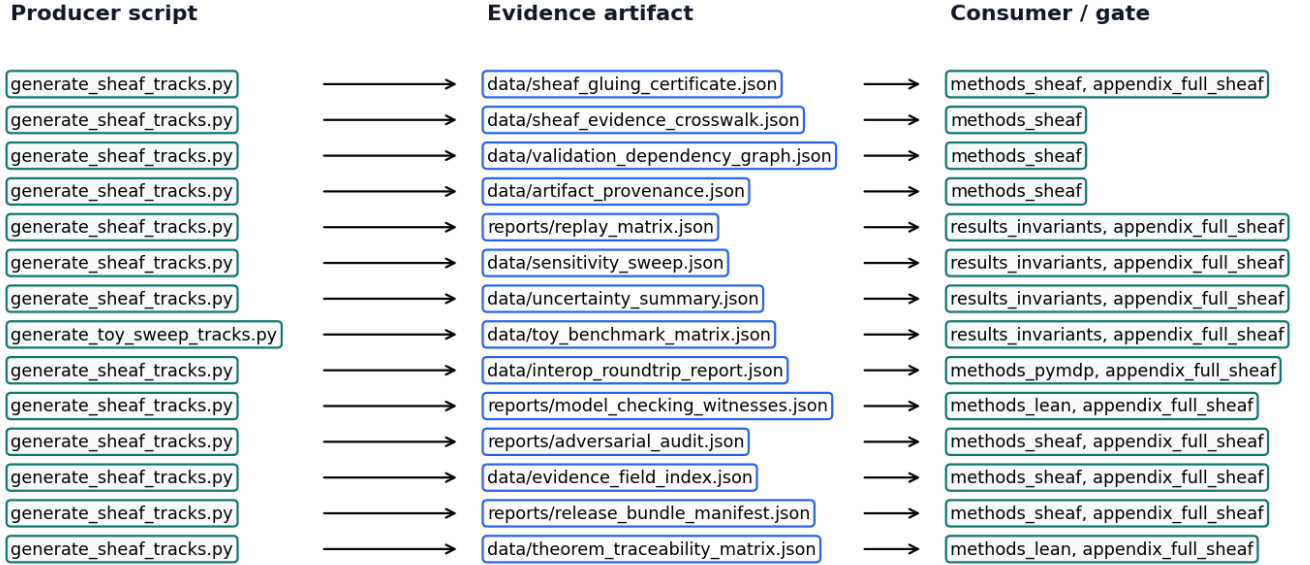


Figure 10: Semantic gluing graph: configured producers, generated evidence artifacts, and validation consumers for the multi-track sheaf certificate.

The `adversarial_audit` fragment makes expected failures part of the sheaf rather than an informal test note. `output/reports/adversarial_audit.json` records 22 known-bad rows and 0 known-good rows passing; publication proceeds only when every row is documented as an expected failure and mapped to a gate.

The audit rows target the same failure modes as the semantic certificate: incomplete sweep cells, unnormalized uncertainty rows, interop field loss, stale certificate state, and empirical-scope leakage. The scope boundary remains toy-only: `toy_only_pass`.

The `evidence_fields` fragment indexes the exact artifact fields that support typed claims and hydrated manuscript tokens. `output/data/evidence_field_index.json` records 81 field rows, and the track passes only when every referenced JSONPath or dotted field is present (`true`).

The `release_bundle` fragment records whether the canonical deliverables exist before copying and whether copied root outputs match or are explicitly deferred until the copy stage. `output/reports/release_bundle_manifest.json` tracks 23 required deliverables with source-present flag `true`.

The `gate_ergonomics` fragment turns validation commands into evidence rows. `output/data/validation_gate_index.json` records 22 gate rows, each naming required inputs and the negative-control surface that should fail closed.

8.4.4 Artifact diffoscope track

The `artifact_diffoscope` track compares saved provenance hashes against live artifact hashes at the artifact root JSONPath. Its proof artifact is `output/reports/artifact_diffoscope.json`: it currently records 37 comparison rows, with equality status `true`.

8.4.5 Artifact license track

The `artifact_license` track classifies generated and project-source artifacts under the public project license boundary. Its audit artifact is `output/reports/artifact_license_audit.json`: it currently records 70 rows, with license-safe status `true`.

The `manuscriptstaleness` fragment closes the hydration loop. `output/reports/manuscriptstaleness_report.json` checks 212 manuscript token bindings against the current generated variables after resolved markdown is written; the pass flag is `true`.

This is a publication-systems claim, not a domain result. A stale hydrated value, unresolved token, or missing resolved section becomes a validation failure before PDF or web outputs are accepted.

8.5 Sheaf fragment track registry

Compose order and renderer bindings from `manuscript/sheaf/tracks.yaml`.

Order	Track id	Label	Renderer	Optional
10	<code>prose</code>	Narrative prose	<code>markdown</code>	No
20	<code>formalism</code>	Mathematical formalism	<code>markdown</code>	No

Order	Track id	Label	Renderer	Optional
30	simulation	Analytical simulation notes	markdown	No
32	assumption_index	Analytical assumption index	markdown	No
35	layers	Sheaf layers tables	layers_report	Yes
40	pymdp	pymdp harness artifacts	markdown	No
41	interop	GNN/ontology/JSON interop checks	markdown	No
42	provenance	Artifact provenance and bundle lineage spine	markdown	No
45	replay_matrix	Deterministic replay matrix	markdown	No
48	counterexample	Expected-failure counterexamples	markdown	No
50	adversarial_audit	Adversarial audit matrix	markdown	No
52	evidence_fields	Evidence field index	markdown	No
53	release_bundle	Release bundle parity manifest	markdown	No
54	gate_ergonomics	Validation gate ergonomics	markdown	No
55	artifact_diffoscope	Artifact diffoscope	markdown	No
56	artifact_license	Artifact license audit	markdown	No
60	sensitivity	Toy sensitivity sweep	markdown	No
62	uncertainty	Toy uncertainty summaries	markdown	No
65	benchmark	Compact toy benchmark matrix	markdown	No
66	manuscript_staleness	Hydrated manuscript staleness report	markdown	No
67	visualization	Figure references	section_figures	No
70	lean	Lean boundary fragment	markdown	No
75	model_checking	Finite-state model checking witnesses	markdown	No
76	theorem_traceability	Lean theorem traceability matrix	markdown	No
77	proof_extraction	Lean proof extraction index	markdown	No
78	state_space_catalog	Finite state-space catalog	markdown	No
79	causal_ablation	Deterministic causal ablation matrix	markdown	No
80	gnn	GNN notation fragment	markdown	No
90	ontology	Active Inference Ontology bindings	ontology_yaml	No
100	animation	Animation fragment	markdown	Yes
102	animation_delta	Animation frame-delta manifest	markdown	No
110	release_notes	Release notes evidence	markdown	No

Track count: 32 registered fragment types.

8.6 IMRAD binding matrix

Section rows versus fragment track columns. **P** = present (bound and file exists); **—** = absent (not bound); **M** = missing (bound, file absent).

[illegible]

Sections	formal	introduction	scope	contributions	bernoulli-ising	pymdp	simulation	harness	lean	formalization	boundary	sheaf	composition	mutual-information	parameter sweep	free-energy	decomposition	t-maze	active-inference	rollout	validation	invariants	limitations and outlook	appendix: full track coverage	total	present	bound	missing	status	release	done
Limitations P and out-look	P	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	P	—	—	P					
Appendix- (group)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Appendix: full track coverage	P	P	—	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P

Totals: 90 present / 90 bound / 0 missing.

Symbol	Coverage color	Meaning
P	Black	Track present (bound and fragment exists)
—	White	Absent (not bound for this section)
M	Gray	Missing (bound but fragment file absent)

8.7 Section-track status

Generated status for the current manuscript sheaf, summarized per composable section.

Section	IMRAD	Bound	Present	Missing	Status
Motivation and scope	introduction	1	1	0	fully_sheafed
Contributions	introduction	3	3	0	fully_sheafed
Bernoulli–Ising analytical model	methods	7	7	0	fully_sheafed
pymdp simulation harness	methods	7	7	0	fully_sheafed
Lean formalization boundary	methods	6	6	0	fully_sheafed
Sheaf composition	methods	13	13	0	fully_sheafed
Mutual-information parameter sweep	results	4	4	0	fully_sheafed
Free-energy decomposition	results	2	2	0	fully_sheafed
T-maze active-inference rollout	results	3	3	0	fully_sheafed
Validation invariants	results	9	9	0	fully_sheafed
Limitations and outlook	discussion	4	4	0	fully_sheafed
Appendix: full track coverage	appendix	31	31	0	fully_sheafed

Section status: 12 / 12 composable sections fully sheafed; 0 required bound fragments missing.

8.8 Track status

Track	Renderer	Bound sections	Present	Missing	Claims	Status
prose	markdown	12	12	0	0	complete

Track	Renderer	Bound sections	Present	Missing	Claims	Status
formalism	markdown	5	5	0	0	complete
simulation	markdown	5	5	0	9	complete
assumption_index	markdown	2	2	0	1	complete
layers	layers_report	1	1	0	1	complete
pymdp	markdown	3	3	0	15	complete
interop	markdown	2	2	0	3	complete
provenance	markdown	2	2	0	12	complete
replay_matrix	markdown	2	2	0	3	complete
counterexample	markdown	2	2	0	2	complete
adversarial_audit	markdown	2	2	0	9	complete
evidence_fields	markdown	2	2	0	1	complete
release_bundle	markdown	2	2	0	4	complete
gate_ergonomics	markdown	2	2	0	4	complete
artifact_di	markdown	2	2	0	1	complete
ffoscope						
artifact_license	markdown	2	2	0	1	complete
sensitivity	markdown	2	2	0	7	complete
uncertainty	markdown	2	2	0	3	complete
benchmark	markdown	2	2	0	3	complete
manuscript_	markdown	2	2	0	1	complete
staleness						
visualization	section_figures	10	10	0	10	complete
lean	markdown	2	2	0	7	complete
model_checking	markdown	2	2	0	6	complete
theorem_tra	markdown	2	2	0	2	complete
ceability						
proof_extraction	markdown	2	2	0	1	complete
state_space	markdown	2	2	0	1	complete
_catalog						
causal_ablation	markdown	2	2	0	1	complete
gnn	markdown	3	3	0	4	complete
ontology	ontology_yaml	5	5	0	5	complete
animation	markdown	1	1	0	2	complete
animation_delta	markdown	1	1	0	1	complete
release_notes	markdown	2	2	0	1	complete

Status cells: 544 section-track cells.

8.9 Render and logging summary

Event	Component	Output	Status	Detail
registry_loaded	sheaf.registry	registered_tracks	ok	32 tracks
manifest_loaded	sheaf.manifest	manifest_sections	ok	17 sections
coverage_matrix_built	sheaf.coverage	output/data/sheaf_coverage_matrix.json	ok	90 present cells
section_status_matrix_built	sheaf.status	output/data/sheaf_section_status_matrix.json	ok	544 section-track cells
layers_renderer_bounded	sheaf.layers_report	manuscript/08_methods_sheaf.md	ok	methods sheaf layer tables
semantic_artifacts_indexed	sheaf.semantic	output/data/validation_dependency_graph.json	ok	70 artifact producer rows
validation_gates_indexed	gates	output/data/validation_gate_index.json	ok	3 gate groups
manuscript_sections_composed	sheaf.compose	manuscript/*.md	ok	16 composed markdown files

Render events: 8.

8.10 Evidence crosswalk

Claim	Artifact	Producer	Gates
sheaf_registry	manuscript/sheaf/tracks.yaml	manual	validate_outputs
sheaf_manifest	manuscript/sheaf/manifest.yaml	manual	validate_outputs
sheaf_coverage_config	manuscript/sheaf/coverage.yaml	manual	validate_outputs
sheaf_coverage_matrix	output/data/sheaf_coverage_matrix.json	generate_figures.py	validate_outputs, validate_manuscript
sheaf_gluing_certificate	output/data/sheaf_gluing_certificate.json	generate_sheaf_tracks.py	validate_manuscript, validate_outputs
sheaf_evidence_crosswalk	output/data/sheaf_evidence_crosswalk.json	generate_sheaf_tracks.py	validate_manuscript, validate_outputs
evidence_fields_mapped	output/data/evidence_field_index.json	generate_sheaf_tracks.py	validate_outputs, validate_manuscript
validation_dependency_graph	output/data/validation_dependency_graph.json	generate_sheaf_tracks.py	validate_manuscript, validate_outputs

Claim rows: 81 typed evidence claims.

8.11 Artifact producer graph

Artifact	Producer	Configured	Consumers
output/data/analysis_statistics.json	compute_statistics.py	Yes	results_si_tmaze, results_invariants
output/data/analytical_assumption_index.json	generate_toy_sweep_tracks.py	Yes	methods_analytical, appendix_full_sheaf
output/data/analytical_observable_sweep.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/animation_frame_deltas.json	render_animation.py	Yes	appendix_full_sheaf
output/data/artifact_provenance.json	generate_sheaf_tracks.py	Yes	methods_sheaf
output/data/causal_ablation_matrix.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/cross_track_symbol_table.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/evidence_field_index.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/figure_source_map.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/gnn_roundtrip_report.json	generate_formal_interop_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/interop_roundtrip_report.json	generate_sheaf_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/manuscript_evidence_tables.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/manuscript_token_provenance.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/manuscript_variables.json	z_generate_manuscript_variables.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/ontology_alias_index.json	generate_formal_interop_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/ontology_profile_matrix.json	generate_formal_interop_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/parameter_sweep.csv	run_analytical_sweep.py	Yes	methods_analytical, results_mi_sweep
output/data/proof_extraction_index.json	generate_formal_interop_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/data/pymdp_policy_posterior_grid.json	simulate_si_tmaze.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/sensitivity_sweep.json	generate_sheaf_tracks.py	Yes	results_invariants, appendix_full_sheaf

Artifact	Producer	Configured	Consumers
output/data/sheaf_coverage_matrix.json	generate_figures.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/sheaf_evidence_crosswalk.json	generate_sheaf_tracks.py	Yes	methods_sheaf
output/data/sheaf_gluing_certificate.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/sheaf_section_status_matrix.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/si_efe_terms.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/si_graph_world_summary.json	simulate_si_graph_world.py	Yes	methods_pymdp, results_si_tmaze
output/data/si_graph_world_topology_sweep.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/si_graph_world_topology_traces.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/si_graph_world_trace.json	simulate_si_graph_world.py	Yes	methods_pymdp, results_si_tmaze, appendix_full_sheaf
output/data/si_policy_comparison.json	simulate_si_tmaze.py	Yes	methods_pymdp, results_si_tmaze
output/data/si_policy_grid.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/si_tmaze_summary.json	simulate_si_tmaze.py	Yes	methods_pymdp, results_si_tmaze
output/data/si_tmaze_trace.json	simulate_si_tmaze.py	Yes	methods_pymdp, results_si_tmaze
output/data/state_space_catalog.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/theorem_traceability_matrix.json	generate_sheaf_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/data/toy_benchmark_matrix.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/track_improvement_scope.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/uncertainty_summary.json	generate_sheaf_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/validation_dependency_graph.json	generate_sheaf_tracks.py	Yes	methods_sheaf
output/data/validation_gate_index.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
../figures/si_belief_trajectory.gif	render_animation.py	Yes	appendix_full_sheaf
output/reports/adversarial_audit.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/artifact_diffoscope.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/artifact_license_audit.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/blocked_scope_manifest.json	generate_sheaf_tracks.py	Yes	methods_sheaf, discussion_outlook, appendix_full_sheaf
output/reports/claim_evidence_audit.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/counterexample_matrix.json	generate_sheaf_tracks.py	Yes	methods_sheaf
output/reports/figure_hash_manifest.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/gnn_lint_report.json	generate_formal_interop_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/reports/graph_world_invariants.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/reports/invariants.json	run_analytical_sweep.py	Yes	results_invariants

Artifact	Producer	Configured	Consumers
output/reports/lean_graph_world_inventory.json	generate_formal_interop_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/reports/lean_theorem_inventory.json	generate_formal_interop_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/reports/manuscript_staleness_report.json	z_generate_manuscript_variables.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/model_checking_witnesses.json	generate_sheaf_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/reports/producer_completeness.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/pymdp_runtime_diagnostics.json	simulate_si_tmaze.py	Yes	methods_pymdp, appendix_full_sheaf
output/reports/release_bundle_manifest.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/release_notes_evidence.json	generate_integration_audit.py	Yes	discussion_outlook, appendix_full_sheaf
output/reports/replay_matrix.json	generate_sheaf_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/reports/reproducibility_replay.json	generate_validation_spine.py	Yes	results_invariants
output/reports/scope_boundary_audit.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/sheaf_renderer_log.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/si_invariants.json	simulate_si_tmaze.py	Yes	results_si_tmaze
output/reports/si_tmaze_runtime_report.json	simulate_si_tmaze.py	Yes	results_si_tmaze
output/reports/stale_artifact_report.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf

Producer issues: 0.

8.12 Semantic gluing restrictions

Restriction	Value
Coverage missing	0
Policy comparison rows	4
Policy grid complete	True
Policy posterior rows	10
Policy posterior normalized	True
Runtime unexpected warnings	0
Graph-world trace agrees	True
Animation frames	4
Lean all proved	True
GNN ontology ok	True
Configured producers ok	True
Semantic certificate ok	None
Dependency edges ok	True
Track scope complete	True
Empirical adapter blocked	True
Provenance bundles complete	True
Replay rows matched	True
Sensitivity complete	True
Uncertainty normalized	True
Evidence fields mapped	True
Release bundle sources present	True
Theorem traceability linked	True
Gate ergonomics indexed	True
Interop lossless	True
Scope toy-only	True

8.13 Track improvement scope

Track	Status	Current proof	Next artifact	Gate	Negative control
adversarial_audit	live	output/reports/adversarial_audit.json	output/reports/adversarial_audit.json	validate_outputs, validate_manuscript	adversarial_known_bad_packages
animation	optional	../figures/sibelief_trajectory.gif	../figures/sibelief_trajectory.gif	validate_outputs	missing_fragment_coverage
animation_delta	live	output/data/animation_frame_deltas.json	output/data/animation_frame_deltas.json	validate_outputs, validate_manuscript	missing_fragment_coverage
artifact_diffoscope	live	output/reports/artifact_diffoscope.json	output/reports/artifact_diffoscope.json	validate_outputs, validate_manuscript	artifact_diffoscope_missed_packages
artifact_license	live	output/reports/artifact_license_audit.json	output/reports/artifact_license_audit.json	validate_outputs, validate_manuscript	artifact_license_unsafe_packages
assumption_index	live	output/data/analytical_assumption_index.json	output/data/analytical_assumption_index.json	validate_outputs, validate_manuscript	missing_fragment_coverage
benchmark	live	output/data/toy_benchmark_matrix.json	output/data/toy_benchmark_matrix.json	validate_outputs	missing_fragment_coverage
causal_ablation	live	output/data/causal_ablation_matrix.json	output/data/causal_ablation_matrix.json	validate_outputs, validate_manuscript	causal_ablation_missing_packages
counterexample	live	output/reports/counterexample_matrix.json	output/reports/counterexample_matrix.json	validate_outputs, validate_manuscript	known_bad_counterexamples
evidence_fields	live	output/data/evidence_field_index.json	output/data/evidence_field_index.json	validate_outputs, validate_manuscript	missing_typed_claim
formalism	live	manuscript/sheaf/manifest.yaml	manuscript/sheaf/manifest.yaml	validate_manuscript	missing_fragment_coverage
gate_ergonomics	live	output/data/validation_gate_index.json	output/data/validation_gate_index.json	validate_outputs, validate_manuscript	gate_ergonomics_unindexed_packages

Improvement rows: 37.

Results

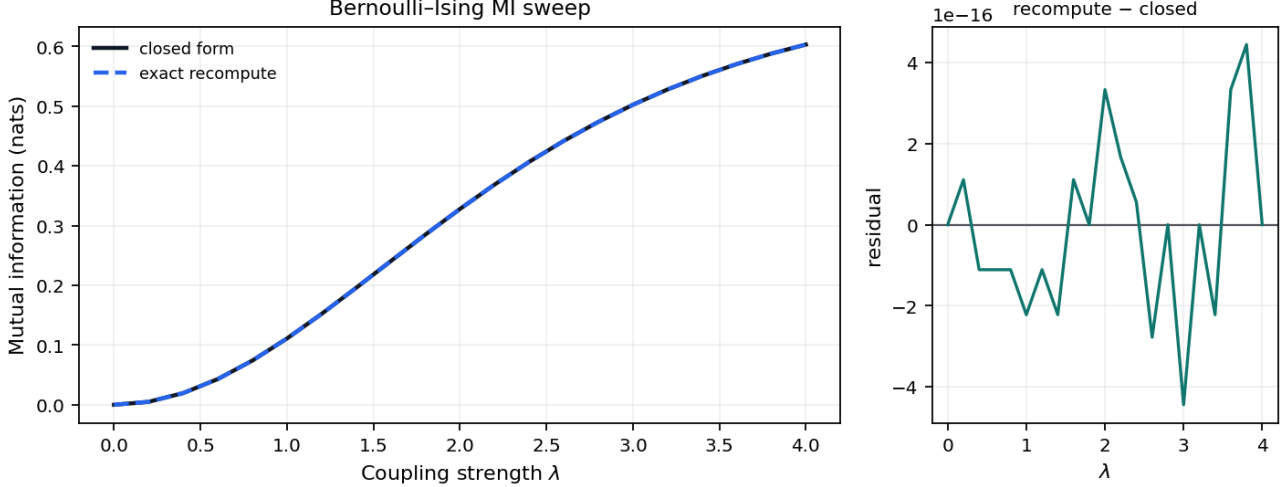
9 Mutual-information parameter sweep

We sweep coupling strength λ on a grid of 21 points up to $\lambda_{\max} = 4$. Closed-form mutual information from eq. 1 is cross-checked against an independent exact recomputation via total correlation from the analytical module (sec. 5); both are deterministic (no sampling) and agree to 0 nats.

Measured invariant checks: 12 / 12 passed on the clean tree.

The sweep reuses the entangled joint defined in eq. 1 (sec. 5). Mutual information $I(\lambda) = \log 2 - H_b(\sigma(\lambda))$ is evaluated on the same λ grid as the analytical oracle and its independent exact recomputation.

Both estimators are deterministic (no sampling, no RNG) and are evaluated on the same λ grid as the closed-form sweep (sec. 5, fig. 5).



Reproduced from fig. 5. Closed-form $I(\lambda)$ and an independent exact recomputation via total correlation for the symmetric Bernoulli-Ising toy across 21 grid points up to $\lambda_{\max} = 4$; grid maximum 0.6031 nats. Both estimators are deterministic (no sampling), so the right panel is a cross-implementation agreement check (max residual 0 nats), not a sampling residual.

10 Free-energy decomposition

Free energy against the entangled prior is evaluated along the same λ grid used for the MI sweep (fig. 11). Against the *entangled* prior the entangled posterior is the exact variational minimizer, so its free energy is identically zero; the Theorem-5.1 decomposition then splits that zero into per-stream marginal free energies, a coupling-cost term, a coupling-prior term, and a total-correlation gain. For the symmetric toy with uniform marginals the coupling-prior term equals $-I(\lambda)$ and exactly cancels the total-correlation gain $+I(\lambda)$ — an exact cancellation the merged invariant suite checks (12/12 pass). The curve in fig. 11 instead reports free energy against the *mean-field* prior: its minimum at $\lambda = 0$ is where the entangled posterior coincides with the factorized mean-field product, and any $\lambda > 0$ raises the free energy as coupling pulls the posterior away from that independent prior.

Saturation MI (grid maximum on the measured λ sweep): 0.6031 nats.

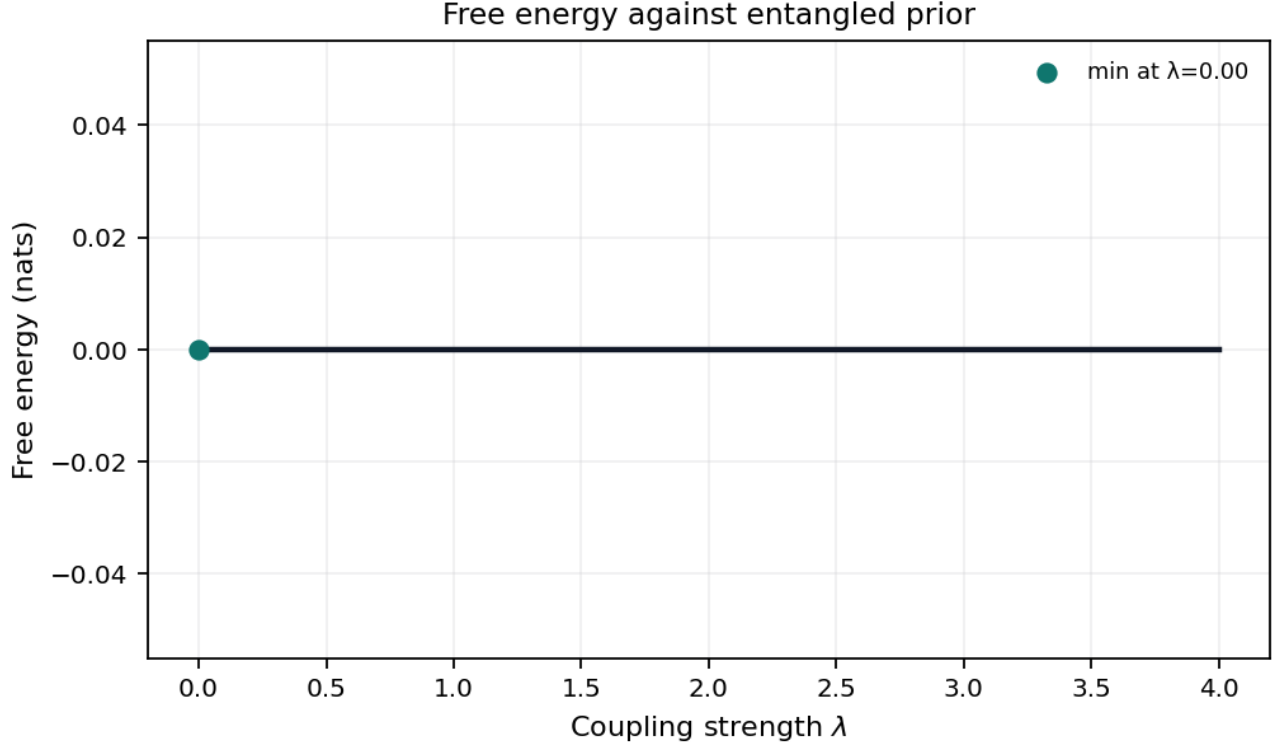


Figure 11: Free energy of the entangled posterior relative to the mean-field prior across the hyperparameter sweep (grid points 21).

11 T-maze active-inference rollout

The pymdp harness rolls out a T-maze active-inference agent in `state_inference` mode with planning horizon 2. The default `state_inference` mode is belief filtering with a goal-seeking action rule; sophisticated policy inference (an expected-free-energy policy posterior) is selectable via `mode: policy_inference` (sec. 6). Summary metrics land in `output/data/si_tmaze_summary.json`.

Steps recorded: 2. Mean belief entropy: 0.3251. Belief entropy over the rollout is traced in fig. 12; the paired observation and action indices are in fig. 13. The default `state_inference` mode runs pymdp `infer_states` and `reports` the resulting posterior (belief entropy and the state-1 marginal), but the action is chosen by an open-loop scripted rule on the observation index — not by the posterior — so the inferred belief here is observed, not acted on. Under the toy transition model, expected-free-energy policy inference reaches the goal in 1 of its rows versus 2 for the scripted state-inference rule: no behavioral advantage on this two-state, horizon-2 maze, which is the measured content of the deliberately-too-small claim.

Policy-comparison rows: 4 across state-inference and policy-inference modes; goal-reaching rows: 3. Graph-world extension rows: 4 over 4 nodes, with goal-reached flag 1.

Rollout trace: `output/data/si_tmaze_trace.json`. JSONL run log: `output/logs/pymdp_runs.jsonl`.

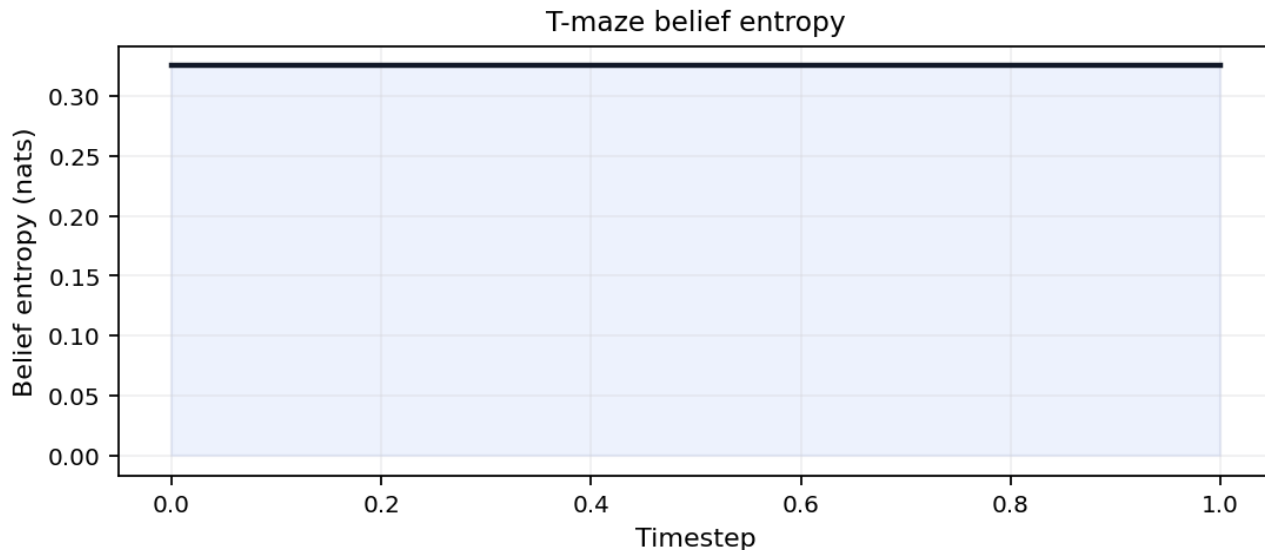


Figure 12: Belief entropy over time for the T-maze rollout (mean 0.3251 nats).

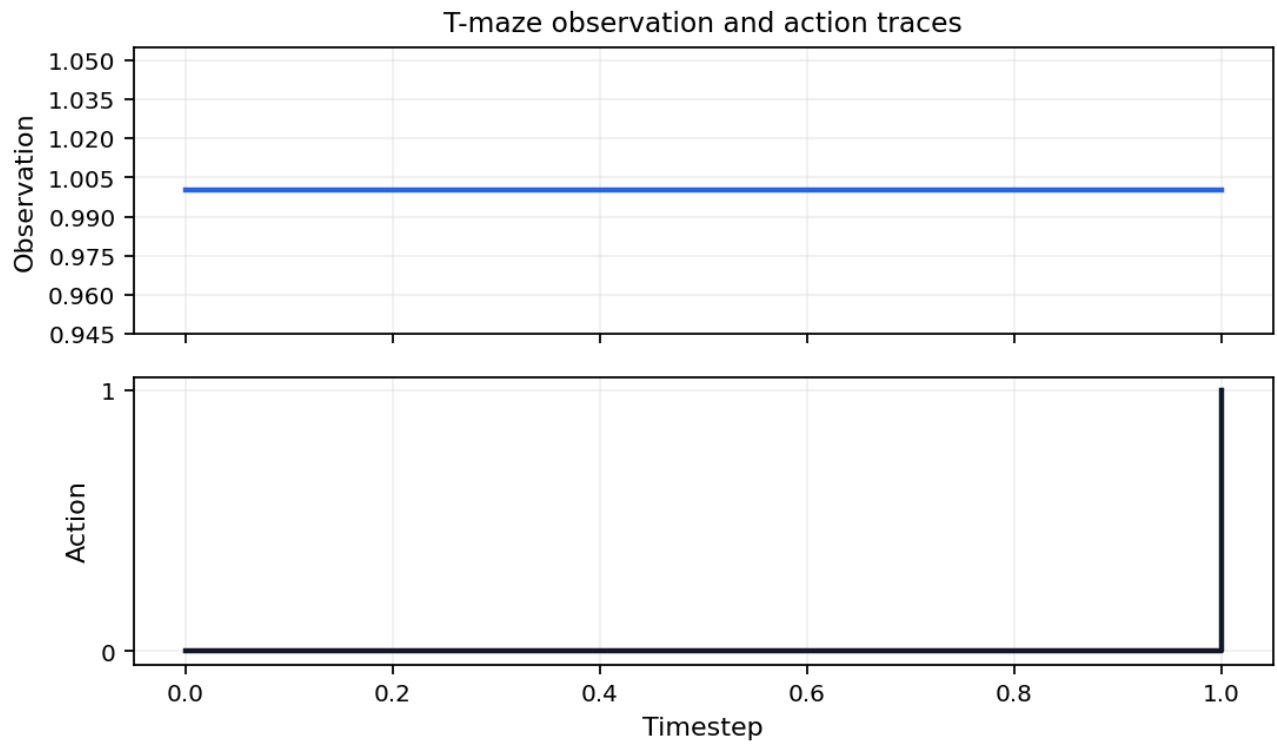


Figure 13: Observation and action traces for the T-maze rollout (action diversity 2).

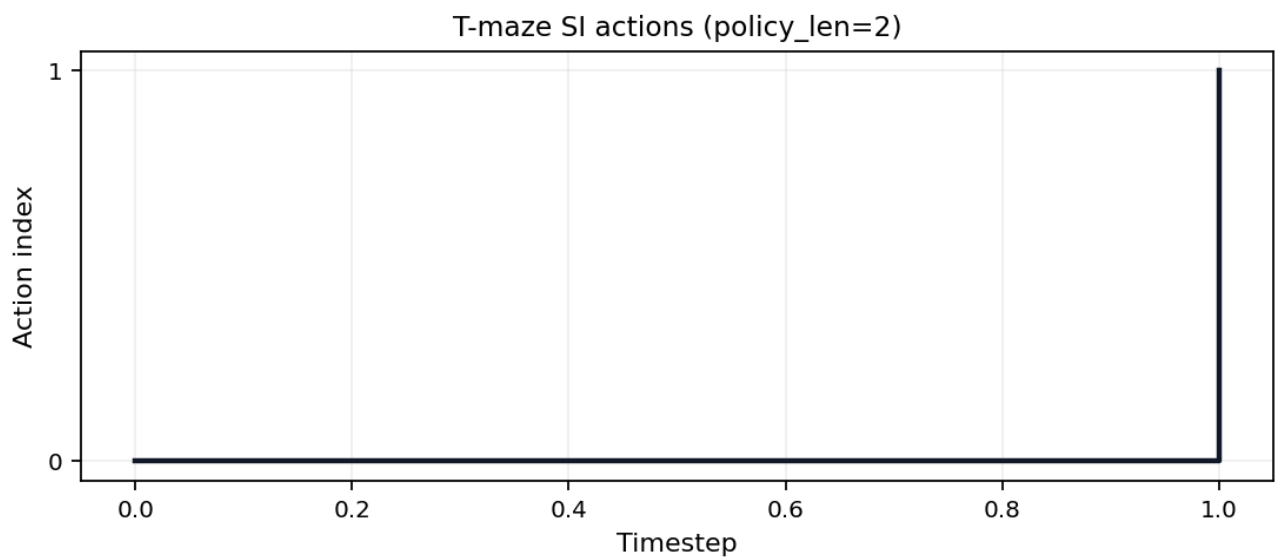


Figure 14: Discrete action index over time for the pymdp T-maze rollout (policy length 2).

12 Validation invariants

The analytical invariant registry runs before PDF rendering (sec. 5). On a clean checkout **12 / 12** checks pass in the merged validation report, which records simulation invariants when the pymdp harness ran (sec. 11).

fig. 15 lists each analytical and simulation gate; failures block publication artifacts. See sec. 8 for how invariant counts hydrate manuscript tokens.

Simulation invariants merge into the analytical report after the pymdp harness runs (sec. 11). fig. 15 summarizes pass/fail status for both domains on the clean tree.

The replay matrix exposes deterministic rerun comparison as table data rather than prose. It contains 13 producer rows, uses explicit replay-or-fingerprint methods, and every row must match its saved artifact hash (**true**).

The **sensitivity** fragment binds the deterministic toy sweep to the canonical sheaf track. **output/data/sensitivity_sweep.json** contains 96 cells across toy parameters, policy modes, seeds, horizons, and graph topologies; the hydrated flag **true** is the only manuscript claim about coverage.

The companion **output/data/si_policy_grid.json** records measured policy-mode rows derived from **si_policy_comparison.json**, not a synthetic grid. Missing cells fail the artifact schema before they can become prose; the topology trace artifact contributes 4 deterministic topology traces.

The **uncertainty** fragment reports only normalized toy summaries. **output/data/uncertainty_summary.json** contains 12 belief and policy-posterior rows plus 3 finite entropy bins, and **true** is false if any posterior row fails to sum to one within the deterministic tolerance.

Policy uncertainty is recorded in generated policy artifacts rather than hand-entered into the manuscript. The posterior grid contributes 5 available posterior rows; the EFE artifact reports availability-or-fallback flag 1. The fragment is therefore a validation surface, not an empirical uncertainty claim.

The **benchmark** fragment adds a compact toy matrix over the Bernoulli, T-maze, and graph-world artifacts. **output/data/toy_benchmark_matrix.json** reports 3 model rows and **true** only when each row names an artifact, metric, and passing gate.

The matrix is scoped to deterministic exemplar models. It is useful as a cross-track smoke test, not as a performance benchmark for biological or deployed systems.

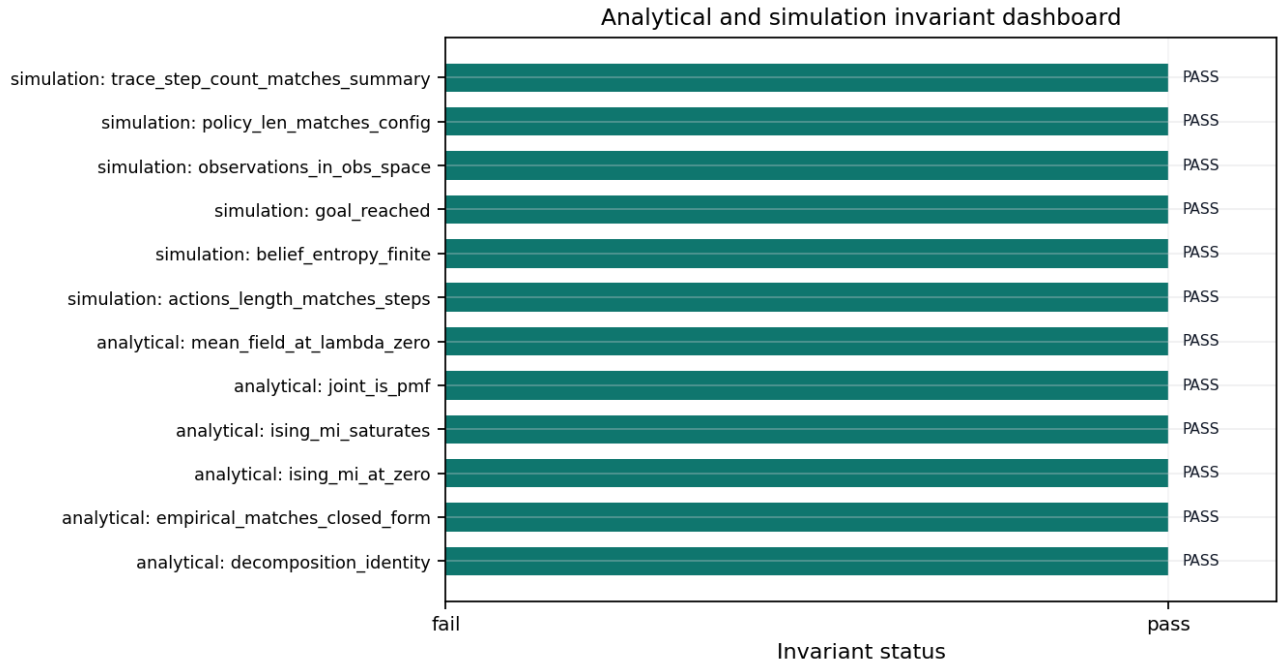


Figure 15: Invariant dashboard: 12 / 12 merged analytical and simulation checks from the validation registry.

12.0.1 State-space catalog track

The **state_space_catalog** track enumerates finite state spaces, action spaces, and policy counts for the deterministic toy models. The catalog artifact is **output/data/state_space_catalog.json**: it currently records 6 rows, with finite-space status **true**.

12.0.2 Causal ablation track

The **causal_ablation** track records deterministic toy ablations over finite preferences, likelihood-noise settings, and graph-topology perturbations. The matrix artifact is **output/data/causal_ablation_matrix.json**: it currently records 36 cells, with complete-grid status **true**.

Discussion

13 Limitations and outlook

13.1 What this demonstrates

The result of this manuscript is a *discipline*, not a domain claim: across three toy models every reported number is hydrated from a generated artifact, 6 sheaf axioms are machine-checked before composition, and 22 negative controls keep each failure path live. No statistic, figure, or cross-track claim here can drift from its artifact without failing a gate before the PDF is built.

13.2 Limitations

The Bernoulli–Ising toy, T-maze harness, and sheaf composition model are pedagogical. They validate analytical consistency, artifact wiring, renderer dispatch, and manuscript hydration—not empirical claims about biological agents. Default pymdp mode is `state_inference` with planning horizon 2; the policy-comparison artifact exposes policy-inference rows without changing the default rollout (sec. 6).

13.3 Sheaf audit and outlook

sec. 1 and sec. 14 make binding state auditable under strict compose validation (sec. 8). Pipeline extensions in `tracks.yaml` `extension_tracks` now write deterministic artifacts: a belief GIF via `render_animation.py` and graph-world SI summary/trace via `simulate_si_graph_world.py`. The appendix row already binds an `animation` sheaf fragment without new manifest rows.

Sweep RMSE 0 nats and SI goal reached 1 summarize measured agreement on the declared grids and rollout. Future work includes full expected-free-energy policy selection, richer graph-world rollouts, and expanded Lean proofs beyond the boundary witnesses in sec. 7.

The discussion ontology binds `coverage_semantics` to the audit matrix in sec. 1, `pedagogical_scope` to the non-empirical scope of the toy models, and `state_inference_mode` to the pymdp harness contract in sec. 6.

Measured pymdp rollout (`state_inference`, config hash 81eb061f43b7bfd7): mean belief entropy 0.3251 nats over 2 steps; goal reached flag 1; action diversity 2.

Analytical sweep residual RMSE 0 nats (max residual 0). Coverage audit: 90 present / 90 bound / 0 missing cells on the IMRAD matrix.

13.3.1 Ontology bindings

- `coverage_semantics` → **Coverage matrix semantics**
- `pedagogical_scope` → **Pedagogical scope**
- `state_inference_mode` → **State inference harness**

13.3.2 Release notes evidence track

The `release_notes` track keeps release-language claims source-backed by validation, semantic, and bundle artifacts. Its evidence artifact is `output/reports/release_notes_evidence.json`: it currently records 3 rows, with source-backed status `true`.

Appendix

14 Appendix: full track coverage

This section is the **composability proof** for the manifest-indexed sheaf model: all 31 appendix-bound fragment tracks render into one flat manuscript section without section-specific compose branches. The registry defines 32 composable types; optional **layers** is methods-only and excluded from this row. The **animation** fragment is bound here as an optional registry type alongside the live proof, simulation, formal, notation, validation-spine, integration, audit, finite-catalog, ablation, license, release-evidence, assumption-index, delta, and staleness tracks.

The proof is a publication-systems check (eq. 4). It demonstrates that heterogeneous fragments share one registry, manifest, renderer dispatch path, coverage matrix, and hydration boundary; it does not assert that every track carries equal scientific weight.

For each track $t \in \mathcal{T}_{\text{Full}}$, the appendix row binds a fragment path $f(t)$ and the composer emits `<!-- sheaf-track:t -->` before the rendered body. Generated renderers such as **section_figures** and markdown renderers pass through the same **resolve_track_body()** dispatch, so the appendix exercises the common compose interface rather than a bespoke appendix path.

$$|\mathcal{T}_{\text{Full}}| = 31 \quad (4)$$

The fragment registry defines 32 composable track types; optional **layers** is bound on the methods sheaf section only. Optional **animation** is bound in this appendix proof; the deterministic GIF artifact in **tracks.yaml extension_tracks** is produced by the core analysis DAG and remains separate from this fragment slot.

Because this appendix binds every non-optional appendix track plus optional **animation**, it is the maximal publication stalk of the coverage presheaf and exercises every publication renderer through the common **resolve_track_body()** dispatch. The same compose path is gated by the 6 sheaf laws verified in sec. 8 (6/6 satisfied): the appendix section glues to a unique output (separation), occupies the terminal position of the linear extension under its own **appendix** group row (poset and gluing), binds only well-typed fragments (typing), and owns every fragment path it references (compositionality). No count in this appendix is hand-written; all are injected from the registry-backed oracle.

Analytical sweep artifacts feed sec. 9 and sec. 12; simulation invariants merge after sec. 11. No additional path listing is required beyond those Results sections.

The appendix **assumption_index** row points to **output/data/analytical_assumption_index.json**. It binds the finite Bernoulli-Ising assumptions to equation identifiers and generated artifacts so analytical signposting can be checked mechanically.

pymdp harness summary: **output/data/si_tmaze_summary.json** (mean belief entropy, action trace). Runtime diagnostics: **output/reports/pymdp_runtime_diagnostics.json** (known warnings 4, unexpected warnings 0). Policy posterior grid: **output/data/pymdp_policy_posterior_grid.json** (10 rows). Full log: **output/logs/pymdp_runs.jsonl**.

sheaf-track:interop binds **output/data/interop_roundtrip_report.json**, **output/data/gnn_roundtrip_report.json**, **output/reports/gnn_lint_report.json**, and ontology profile artifacts into the appendix proof row. The appendix claim is exactly 5 checks with lossless status **true**.

The appendix provenance fragment points to **output/data/artifact_provenance.json**, the canonical artifact that records required toy artifact hashes, producer scripts, source commit, deterministic seeds, config digests, and 5 bundle rows.

replay_matrix.json provides the appendix proof for deterministic replay: 13 producer replay/fingerprint rows with matched status **true**.

The appendix counterexample fragment points to **output/reports/counterexample_matrix.json**, the expected-failure matrix that keeps promoted validation gates falsifiable.

sheaf-track:adversarial_audit binds **output/reports/adversarial_audit.json**, **output/reports/scope_boundary_audit.json**, and claim-audit outputs. The appendix claim is exactly 22 expected-failure rows with documented status **true** and known-bad-passing count 0.

evidence_field_index.json provides the appendix proof for field-level claim evidence: 81 mapped fields with status **true**.

release_bundle_manifest.json provides the appendix proof for required deliverables: 23 artifacts with source-present status **true**.

validation_gate_index.json provides the appendix proof for gate ergonomics: 22 indexed gates.

14.0.1 Appendix track: artifact diffoscope

artifact_diffoscope binds **output/reports/artifact_diffoscope.json** into the full sheaf appendix. Rows: 37. All equal: **true**.

14.0.2 Appendix track: artifact license

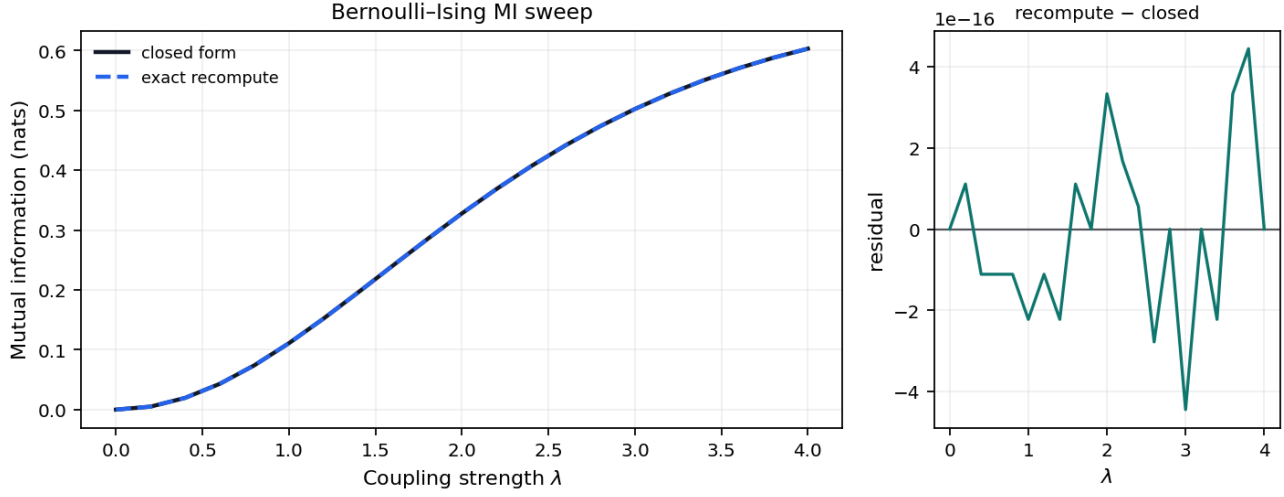
artifact_license binds **output/reports/artifact_license_audit.json** into the full sheaf appendix. Rows: 70. All safe: **true**.

sheaf-track:sensitivity binds **output/data/sensitivity_sweep.json**, measured **output/data/si_policy_grid.json**, **output/data/si_efe_terms.json**, **output/data/analytical_observable_sweep.json**, and graph-world topology artifacts including **output/data/si_graph_world_topology_traces.json**. The appendix claim is exactly 96 complete canonical grid cells.

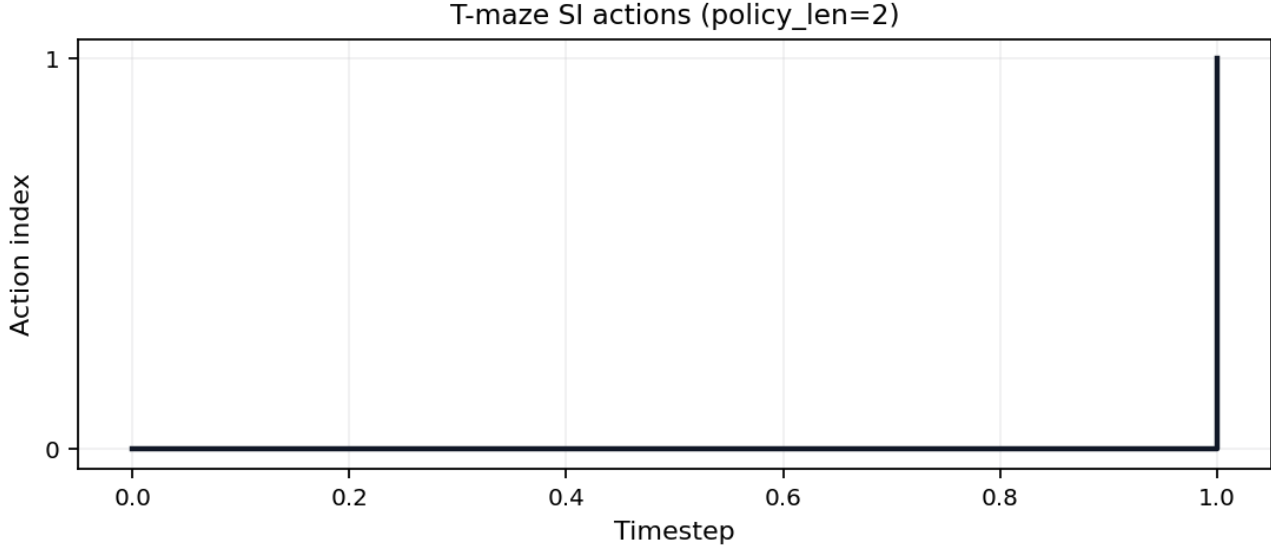
sheaf-track:uncertainty binds **output/data/uncertainty_summary.json**. The appendix claim is exactly 12 normalized rows across 3 entropy bins with status **true**.

sheaf-track:benchmark binds **output/data/toy_benchmark_matrix.json**. The appendix claim is exactly 3 complete toy-model rows with status **true**.

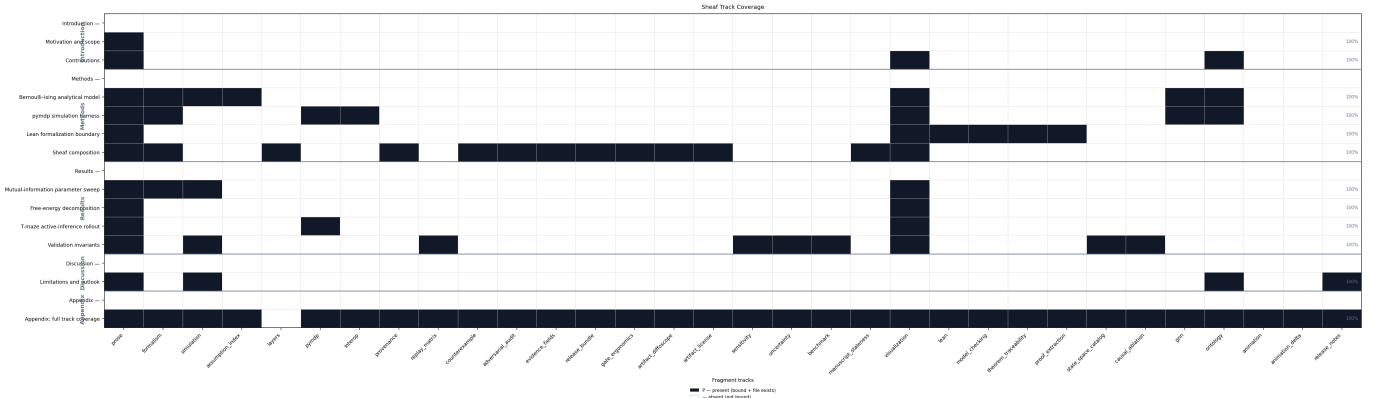
The appendix **manuscript_staleness** row points to **output/reports/manuscript_staleness_report.json**. It checks that generated variables and resolved manuscript markdown agree after hydration, including late audit variables.



Reproduced from fig. 5. Closed-form $I(\lambda)$ and an independent exact recomputation via total correlation for the symmetric Bernoulli-Ising toy across 21 grid points up to $\lambda_{\max} = 4$; grid maximum 0.6031 nats. Both estimators are deterministic (no sampling), so the right panel is a cross-implementation agreement check (max residual 0 nats), not a sampling residual.



Reproduced from fig. 14. Discrete action index over time for the pymdp T-maze rollout (policy length 2).



Reproduced from fig. 3. Sheaf track coverage matrix: 17 IMRAD rows $\times$ 32 fragment columns. Black = present (P), white = absent (—), gray = missing (M). Counts: 90 present / 90 bound / 0 missing.

Lean modules under `lean/TemplateActiveInference/` declare horizon and coupling witnesses. Build with `lake build` in `lean/`; fig. 8 summarizes proved versus deferred statements for this boundary fragment.

`sheaf-track:model_checking` binds `output/reports/model_checking_witnesses.json` and the Lean theorem inventories. The appendix claim is exactly 12 finite exhaustive witnesses with pass status `true`; Lean graph-world topology coverage is 4 generated topology ids with all-witnessed flag `true`.

`theorem_traceability_matrix.json` provides the appendix proof for theorem traceability: 11 linked rows with status `true`.

14.0.3 Appendix track: proof extraction

`proof_extraction` binds `output/data/proof_extraction_index.json` into the full sheaf appendix. Extracted theorems: 10. Constructive status: `true`.

14.0.4 Appendix track: state-space catalog

`state_space_catalog` binds `output/data/state_space_catalog.json` into the full sheaf appendix. Rows: 6. All finite: `true`.

14.0.5 Appendix track: causal ablation

`causal_ablation` binds `output/data/causal_ablation_matrix.json` into the full sheaf appendix. Cells: 36. Complete grid: `true`.

GNN declarations: `gnn/bernoulli_toy.gnn.md` and `gnn/si_tmaze.gnn.md`. [fig. 6](#) and [sec. 5](#) document ontology concordance for the Bernoulli toy; SI notation extends the same pattern under [sec. 6](#).

14.0.6 Ontology bindings

- `belief_entropy` → `BeliefEntropy`
- `expected_free_energy` → `ExpectedFreeEnergy`
- `location` → `HiddenState`
- `observation` → `ObservationLikelihood`
- `policy` → `PolicyPosterior`
- `sheaf_track` → `SheafFragment`

Animation is an **extension** sheaf track backed by a deterministic GIF from `scripts/render_animation.py`. This appendix row documents the track binding only; default publication still uses static SI figures ([sec. 11](#), [fig. 14](#)) while the GIF remains an auditable generated artifact.

The appendix `animation_delta` row points to `output/data/animation_frame_deltas.json`. The manifest records 3 adjacent-frame deltas, with `true` as the hydrated evidence that the GIF is trace-derived rather than a duplicated static frame.

14.0.7 Appendix track: release notes evidence

`release_notes` binds `output/reports/release_notes_evidence.json` into the full sheaf appendix. Rows: 3. Source-backed: `true`.

15 Conclusion

Analytical oracles (sec. 5), pymdp rollouts (sec. 11), and sheaf composition (sec. 8) share one auditable manuscript contract: measured artifacts hydrate 12 composed sections, sec. 1 reports binding state, and strict compose validation blocks gray matrix cells before PDF rendering.

The T-maze harness runs in **state_inference** mode with config hash **81eb061f43b7bfd7**; sweep RMSE 0 nats summarizes analytical–empirical agreement on the toy coupling grid. sec. 12 merges analytical and simulation gates; sec. 13 states scope and extensions. Scientific claims remain confined to declared models—not empirical statements about biological agents.

16 References

See `manuscript/references.bib` for bibliography entries cited in the composed sections.

References

END OF TRANSMISSION

Release: v0.3.0 · DOI 10.5281/zenodo.20417021 · SHA-256 pending... · pairing pending



Figure 16: Integrity QR strip

Prior: No prior releases.